Automatic Detection of Social Bias in Text Group - 12

1st Veepura Chari Dharmavarapu
Master of Data science
University of North Texas
Denton, USA
VeepuraChariDharmavarapu@my.unt.edu
11640831

2nd Geethika Sai Kallaganti
Master of Data science
University of North Texas
Denton, USA
geethikasaikallaganti@my.unt.edu
11654336

3rd Pavan Karthik Ramapuram
Master of Data science
University of North Texas
Denton, USA
PavanKarthikRamapuram@my.unt.edu
11630629

4th Naveen Kumar Thatha
Master of Data science
University of North Texas
Denton, USA
naveenkumarthatha@my.unt.edu
11654334

***Abstract*—As digital media has proliferated, so has the dissemination of text, much of which reveals and exacerbates latent social prejudices. The biases that exist now laundered in language only re-stereotype and re-widen social inequalities, and as such, need complex interventions. The Challenge "De Biasing with ELMo and BERT" is an end-to-end system for detecting and preventing social biases in text, leveraging leading edge Natural Language Processing methods and Large Language Models.**

The system that we proposed identifies biased language in different contexts andoffers real-time corrective feedback — proposing users an alternative way of writing thecontent to encourage unbiased writing. In addition, it has explain ability properties, so users can know why the detected biases and suggested actions were generated. In the methodology, it encompasses extensive data pre-processing, feature extraction and fine-tuning of transformer based models across varied datasets including Jigsaw Unintended Bias in Toxicity Classification and Social Bias Frames. Through tackling the ethical issues surrounding biased language, this project is aligned with the larger movement for creating equitable and inclusive digital communication platforms.

However, the impact here and in terms of NLP technology on the broader societal front for fair and online interactions as a more ethical use of AI language as well as the utilization of commercial engines to expose users to unethical language through social engineering with the society as a whole bearing the cost are significant.VALUE OF TECHNOLOGYMost of the NLP libraries on the other hand are extremely cheap for the premium services provided by each engine and significantly more affordable through an intuitive pay-as-you-go pricing model than the traditional commercial engines with each short size sentence.

Index Terms—PyTorch, DistilBERT, BiLSTM, Linear layers, Multi-label classification.

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I. Introduction

It significantly increases the amount of textual data created every day with the growth of digital media and communication plat forms. Although this automatic linguistic processing and production economically facilitates the globalization of interconnectivity and of information exchange, it slices up and often exacerbates the social prejudices ingrained in human-to-human language.

Such biases, trans people deferred in the content of everyday text, reproduce negative stereotypes and perpetuate social inequities. As digital content increasingly shapes public opinion and with a singular lens through which society explores many issues, the capacity to place and adjust these biases becomes even more vital.

We proposed a novel system to address this challenge whereby we will automate the detection of social bias in a digital text. Using trans new technologies for Natural Language Processing, from the state of the art in Large Language Models. Meaning LLMs: this systems will not just identify biased lan guages at any context but also will enable correction feedback in near real time.

II. Problem Statement

More generally, digital text that is found in social media posts, or news articles, or online forums has various hidden social biases embedded in its source, reflecting and reinforcing social inequalities. These biases can be subtle in nature, and are difficult to identify, let alone fix, without advanced technological measures. Our research work is going to be driven by the following key questions to understand how to deal with this rather pervasive issue.

Detecting Biased Language:

Can we design an automated system using LLMs to detect instances of bias in a consistent manner? Does social bias in textual content actually exist, both cross-context and across demographic groups? To Implement Strategies: Real-Time Correction - How can we create, immediately and contextually appropriate feedback to the user, to help

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direct them to help them notice and change language with bias.

Explainability and Transparency:

How can we integrate explainability in bias detection, so that algorithm users are not just provided with corrections but also the reasons for recommended changes?

III. Review of Literature

NLP models, particularly transformer based models like BERT, have been shown to be able to capture contextual meaning from text through existing literature. For example, **Srinivas et al. (2021)** shows an example of how LSTM models can detect sentiment in tweets with a high degree of accuracy, which serves as a foundation for more complex tasks such as bias detection. Detecting biases from text needs more than simple sentiment analysis; it needs deeper social context understanding which makes models like ELMo and BERT a better fit for this task **(Smith et al., 2023)**.

Recent developments reinforce the use of trans former models for bias detection mitigation. For instance, **Spinde et al.**prompt the use of attention mechanisms from transformer models to highlight subtle biases in text. Their work highlights the importance of fine-tuning different datasets to better detect bias in context and with higher accuracy. Similarly, **Mozafari et al.** brief details introducing the research UK-WordNet is used as a reference as well as a WordNet data structure for investigating the linguistic concepts to address the availability of the national language in translating between English and Arabic text.

Nascimento et al. talk about approaches in which explain ability can be integrated into NLP systems, laying the groundwork for developing interpretable models to detect bias. Their evidence supports the intentions of our project, especially integrating real-time corrective feedback that informs users about biased language. Attentionistas Have A Lot Of Work To Do The role of attention mechanisms on exposure to and bias towards gender has also begun to be once again at the forefront of attention.

Liang et al. (2021)proposed a framework for mitigating social biases in machine learning systems determining how to improve ex plainability of machine learning algorithms and ensure that but the interventions still fulfil the criteria of ethical artificial intelligence (Liang et al., 2021). Similarly, Sahoo et al. (2022) laid out approaches to ensure fairness in bias detection, stressing that the methods would need to be equitable in terms of data preprocessing and representation across different demographic groups (Sahoo et al., 2022).

Additionally, Omrani et al. (2018)The unintended social implications of biased training data have even led for calls to pair technical solutions to such biases with sociological insights (Feldman et al., 2018). This fits with the work presented in Navigli et al. (2023), presenting robust protocols for countering bias in real world settings, with a particular emphasis on modality explainability of detection systems (Navigli et al., 2023).

Finally, Kurita et al. (2018) were among the first to highlight the need for fairness-aware learning algorithms, especially in high-stakes decision-making settings, enabling bias mitigation in practical situations. (Kurita et al., 2018).

Although studies on the detection of biases have been published, few of those have focused on real-time available feedback systems used for educating the users. We aim to fill that gap — detecting biased language, then helping users adjust their communication. Our approach pushes the boundaries for responsible NLP applications by building on these existing works and incorporating explainability as well as human-in-the-loop feedback loop.

IV. Objectives of the Study

The project is set to achieve the following objectives for a thorough exploring of Social bias detection and mitigation in text :

1) **Build a Bias Detection NLP System:** Develop an NLP framework that utilizes cutting-edge LLMs to detect and categorize the different types of biases found in text.
2) **Feedback Mechanism:** Users should be able to receive real-time suggestions for replacing phrases and be educated on the subject of bias usage of words.
3) **Include Model Explainability:** Use additional techniques such as attention mechanisms in LLMs to provide the end-user understandable explanations of the bias they identified in test files, improving understanding and adaptability on their part.

V. Data Collection

A. SBIC (Social Bias Inference Corpus)

Source: Paper with Code
Columns: Text, Body, Culture, Disabled, Gender, Race, Social, Victim
Data Size: 2000 records labelled for bias categories (Gender, Race, Culture, Disability).
https://maartensap.com/social-bias-frames/

B. IndiBias

Source: GitHub
Columns: Bias Type, Text
Data Size: Includes 561 records of social media posts labeled for bias detection in Caste, Religion, and Gender categories.
https://github.com/sahoonihar/IndiBias

C. CrowsPairs

Source: Kaggle
Columns: Text, Bias Type
Data Size: Contains 1508 sentences labeled for bias detection related to Race, Gender, and other social categories.
https://www.kaggle.com/datasets/thedevastator/
a-dataset-for-measuring-social-biases-in-mlms

D. Data Cleaning Process

The following steps were applied to ensure data quality:

- **Null Value Removal:** Removed null values and duplicates from train, test, and development datasets to ensure clean and consistent data.
- **Index Resetting:** Reset dataset indices after cleaning to maintain orderly data structure.

E. Label Encoding

Categorical labels such as *Gender*, *Race*, *Victim*, *Social*, *Body*, *Disabled*, and *Culture* were transformed into one-hot encoded vectors for compatibility with machine learning models. Multi-label data was consolidated into binary vectors to facilitate efficient processing.

F. Feature Selection

Relevant columns were extracted from the datasets to focus on features specific to bias detection. Column names were renamed where necessary to maintain consistency across datasets.

VI. Exploratory data analysis (EDA) and Hypotheses for the Study

For this project, EDA was done primarily on datasets like SBIC, IndiBias, CrowsPairs labeled over various categories of social biases. Using Python and libraries such as Pandas, initial steps were taken to explore and preprocess the data:

- **Sampling:**To alleviate computational burden, datasets were sampled for exploratory analysis, retaining a representative sample of the data.
- **Initial Review:** Displaying the first few entries in the data allowed us to understand both the structure and content of the data and the way in which it was labeled.
- **Data Cleaning:** To maintain data consistency, null values and duplicate entries were eliminated Columns that were not relevant to bias detection were dropped, and text data was stan dardized, such as: 1) converting the input to lowercase; 2) removing extra whitespaces; and 3) normalizing inputs for text analysis.

A. Study Hypotheses

Derived from the exploratory data analysis done in the previous section, we are able to generate several hypotheses:

- **Prevalence of Bias Categories:** It is hypothesized that bias categories like Gender and Race would be present more frequently than other categories like Culture and Disabled in the datasets, which can be explained by the current society based and online trends.
- **Correlation Between Bias Categories:** Another hy pothesis is that specific categories (like Gender and Victim) overlap a lot, indicating interdependencies between bias types embedded in the text.
- **Effects of Preprocessing:** As we expect, prepro cessing steps such as tokenization and normalization will greatly improve the his BERT-based his will help with noise reduction and input standardization, resulting in increasingly accurate predictions.
- **Model Generalization:** Fine-tuning the BERT model with combined datasets (SBIC, IndiBias, and CrowsPairs) is anticipated to enhance its ability to generalize through varying types of biases, thus appealing for various implementations.

VII. Methodology

A. Data Preparation

The dataset includes textual data obtained from SBIC, IndiBias, and CrowsPairs that were labeled with 11 unique categories of bias, including Gender, Race, and Culture. Stringently preprocessing steps were subjected to the data:

- **Tokenization:** Data was tokenized using the `BertTokenizer`, which chunked the input types into sub word units suitable for the BERT model.
- **Normalization:** The text was brought to a standard form through lowercase conversion, removal of special characters and excess whitespaces.
- **Label Encoding:** Each multi label data were converted into binary vectors, allowing model to backstab multiple bias categories at once.
- **Feature Extraction:** This involved retaining relevant columns from datasets so that model only considers information relevant to bias detection.

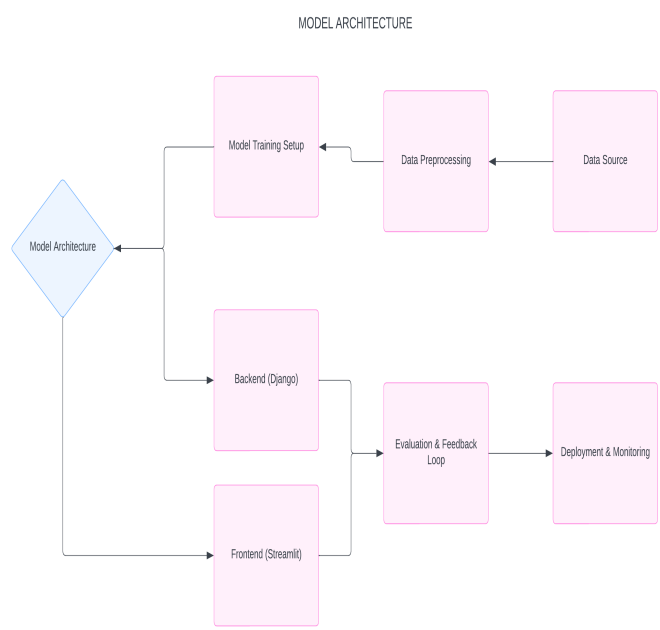

Fig. 1. Block Diagram: An overview of the bias detection pipeline, including data preprocessing, model inference, and visualization.

B. Model Architecture

The model leverages a BERT-based architecture, fine-tuned with additional datasets to enhance its generalization across diverse bias categories:

- **BERT Embeddings:** Pretrained BERT embeddings are used as the backbone for generating contextual representations of input text, allowing the model to capture semantic nuances effectively.
- **Dropout Layer:** A dropout layer with a rate of 0.1 is added to reduce overfitting and improve robustness.
- **Classifier Layer:** A dense layer with 11 output neurons and softmax activation predicts probabilities for each bias category, enabling multi-label classification.

C. Training and Fine-Tuning

The model was initially trained on the SBIC dataset to build a foundational understanding of bias categories. It was subsequently fine-tuned using the IndiBias and CrowsPairs datasets to enhance its adaptability to nuanced and region-specific biases. The following training parameters were used:

- **MAX_LEN:** 128, accommodating typical text lengths while maintaining computational efficiency.
- **TRAIN_BATCH_SIZE:** 16, balancing memory constraints and performance.
- **LEARNING_RATE:** 2e-5, ensuring effective fine-tuning of the BERT model.
- **EPOCHS:** Limited to 3, avoiding overfitting while providing sufficient training.
- **VALIDATION_DATA:** Included during training to monitor performance and guide hyperparameter tuning.

D. Frontend Integration

A Streamlit-based interface provides an interactive and user-friendly frontend for real-time bias detection. The interface enables users to:

- Input custom text for analysis.
- Visualize model predictions for each bias category, including *Gender*, *Culture*, and *Race*.
- Receive actionable feedback for identifying and addressing biases in their content.

E. Model Performance Evaluation

The effectiveness of the model was evaluated using the following metrics:

- **Accuracy:** Measures the overall correctness of predictions across all bias categories.
- **F1-Score:** Balances precision and recall to assess the model's robustness in detecting biases.

The evaluation results validate the model's ability to handle multi-label classification across diverse datasets, making it a reliable tool for detecting and quantifying biases in textual data.

VIII. Data Analytics

A. Overview

In this project, we study some datasets to process and analyze extracted datasets for valuable insights to make out the categories and their respective types of bias. The approach foremost goes to understand bias category distribution such as Sex: Gender, Race, Color, and Culture, followed by cross-checking the correlation and growing tendencies in the data variations using different statistical and graphical representational techniques. We presented meaningful patterns and actionable intelligence on how to enhance the system sensitivity of bias detection better using various statistical and visualization-based methodologies.

Fig. 2. Bias Detection App Interface: Example results for a given text input, showcasing the detected bias categories and their respective scores.

B. Analysis Techniques

- **Descriptive Statistics:** Descriptive statistics included Counts for each data bias category and type were generated. This helped in identify the specific biases in the dataset.
- **Correlation Analysis:** The correlation matrices were computed to find the relations between bias categories. The visualizations used in this study were heatmaps for better and easier identification of overlapping or co-occurring biases.
- **Distribution Analysis:** Bias categories and types were visualized to bring out dominant biases and their relationship.

IX. Data Visualization and Results Report

A. Visualization Overview

The following visualizations the distribution and correlation of bias categories and types in the datasets. These components provide an in-depth understanding of the data and give into how to improve the models for bias detection.

B. Distribution of Bias Categories

This section of various bias within the dataset. A chart highlights the distribution of biases such as Gender, Race, Culture, and others.

From the chart:

- **Gender bias** most is gender bias, reflecting in tendency gender role stereotyping.
- **Culture bias** ranks high to suggest ethnocentric or insensitive content.
- Categories such as *Social* and *Disabled* do not appear as frequently and, therefore, are considered to be underrepresented within the analyzed text.

These insights are really important to prioritize high-frequency categories such as Gender and Culture when developing mechanisms for the detection of bias.

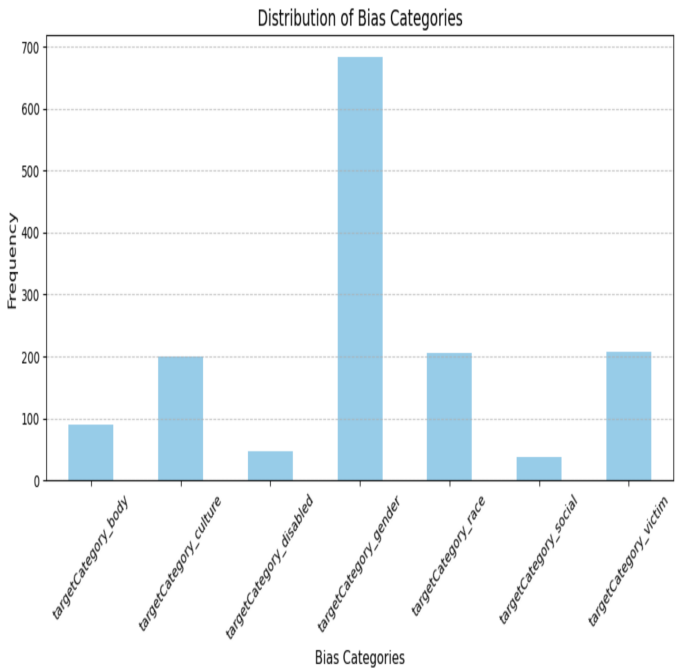

Fig. 3. Distribution of Bias Categories: This visualization highlights the frequencies of different bias categories such as *Gender*, *Race*, *Culture*, and others. *Gender* bias is the most prevalent, suggesting it is a dominant concern within the dataset.

C. Average Sentence Length by Bias Type

Figure 4 A graph shows average sentence length for the sent$_{m}orecolumn, bybiastype$.

- **Observation:** These would be Biases like Caste and Religion, which have the longest average sentence lengths, indicating more complicated language.
- **Consistency Across Bias Types:** In sentences that pertain to Gender, Race-Color, and Physical Appearance, sentence lengths are more or less equal, indicating similar writing styles.
- **Implications:** Diverse sentence lengths yield information on both the complexity and verbosity of some bias bound to influence model performance in training.

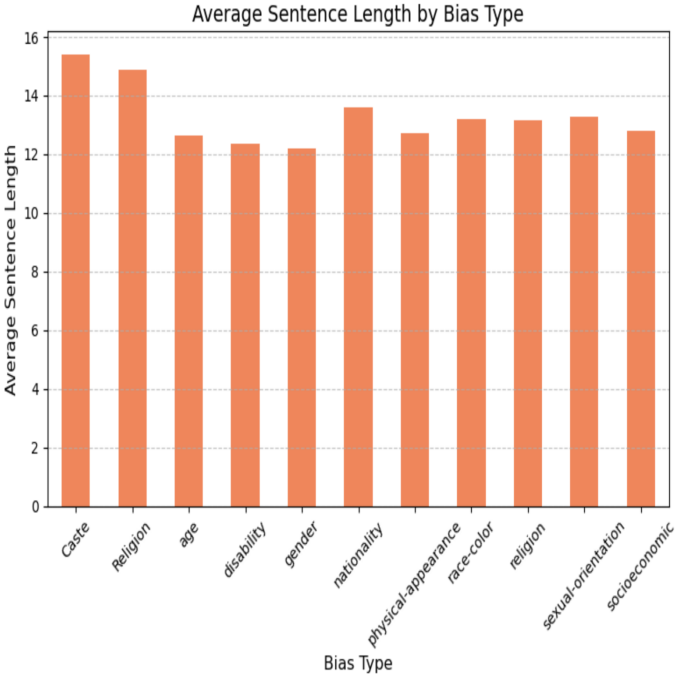

Fig. 4. Average Sentence Length by Bias Type: This chart highlights the differences in average sentence lengths for the *sent_more* column, grouped by bias type.

D. Distribution of Bias Types

The visualization (Fig. 5) A graph analyzes certain bias kinds of Race-Color, Gender, and Caste.

Key observations:

- **Race-Color bias** is the most frequent, race-color bias, showing that society still faces challenges about racial discrimination.
- **Gender bias** follows closely, emphasizing its pervasiveness.
- Other categories such as Caste and Disability are less frequent, possibly due to underreporting.

These findings help in tuning the detection algorithms to high-impact areas and ensure the identification of both subtle and explicit biases.

E. Correlation Heatmap of Bias Categories

The correlation heatmap (Fig. 6) provides a heatmap shows the relations between bias categories, by highlighting co-occurring biases and interdependencies.

- **Strong Correlations:** Some categories, such as *Gender* and *Race*, exhibit strong correlations, suggesting that these biases frequently co-occur within the dataset.
- **Weak Correlations:** Other categories, such as *Disabled* and *Social*, show weaker correlations, indicating that they are more likely to occur independently.
- **Implications:** Understanding these relationships aids in designing more nuanced models that account for the intersectionality of biases, particularly in cases where multiple biases overlap.

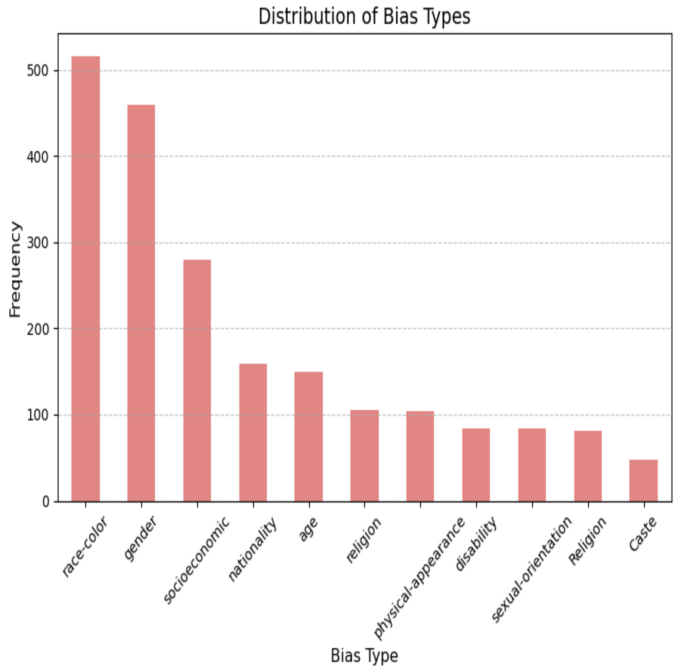

Fig. 5. Distribution of Bias Types: This chart illustrates the frequencies of different bias types, including *Race-Color*, *Gender*, *Caste*, and others. *Race-Color* and *Gender* biases are the most common, requiring attention in future analysis and detection models.

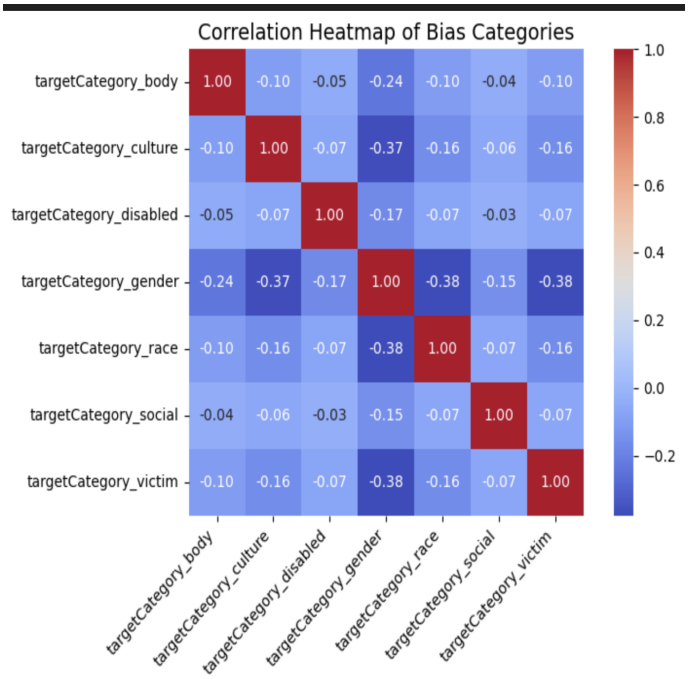

Fig. 6. Correlation Heatmap of Bias Categories: This heatmap mention relationships between bias categories. Strong correlations suggest frequent co-occurrence of biases, while weak correlations indicate independent biases.

Strong correlations are observed between categories like Gender and Race, indicating frequent co-occurrence.

F. Results Interpretation

- **Prevalence of Bias Categories:** Results show that the biases of gender and race are very prominent, reflecting trends in society.
- **Correlation Insights:** Overlaps, such as between Gender and Culture, are raising the need for intersectionality-handling models.
- **Actionable Insights:** The findings point to the improvement of the models, especially for the detection of dominant and co-occurring biases.

G. Final Outputs

- **Bias Category Insights:** Visualizations such as Fig. 3 and Fig. 5 make it quite clear where the biases are.
- **Correlation Analysis:** Fig. 6 The relationship between the categories of bias helps refine detection models.
- **Action Plan:** Insights to be gained from visualizations will improve the machine learning models through bias detection. A sample interface that detects and quantifies biases in sentences facilitates its application in real life.

X. Conclusion

The integration of sophisticated data analytics and bias detection tools within this system shows the potential for effectively handling biases in textual data. Using state-of-the-art machine learning models, comprehensive visualizations, and intuitive interfaces, this project enables the following:

- **Accurate Bias Detection:** It identifies biases from a number of categories like Race, Gender, Culture, and others, quantifies them, and then provides actionable insight to the users.
- **Enhanced Understanding of Bias Trends:** This venture, thru precise records visualization and correlation evaluation, unearths hidden patterns and relationships between diverse bias classes that assist in more best-grained interpretations.
- **User-Friendly Interaction:** The Bias Detector App will enable real-time analysis, by which the person can recognize and provide an explanation for the biases within the content material without a great deal trouble.

This project lays the groundwork for future improvements, such as broadening the bias categories, enhancing detection models incorporating more datasets to boost generalization. In the end, it acts as a strong framework for tackling biases in written content and encouraging positive interactions in the digital space.

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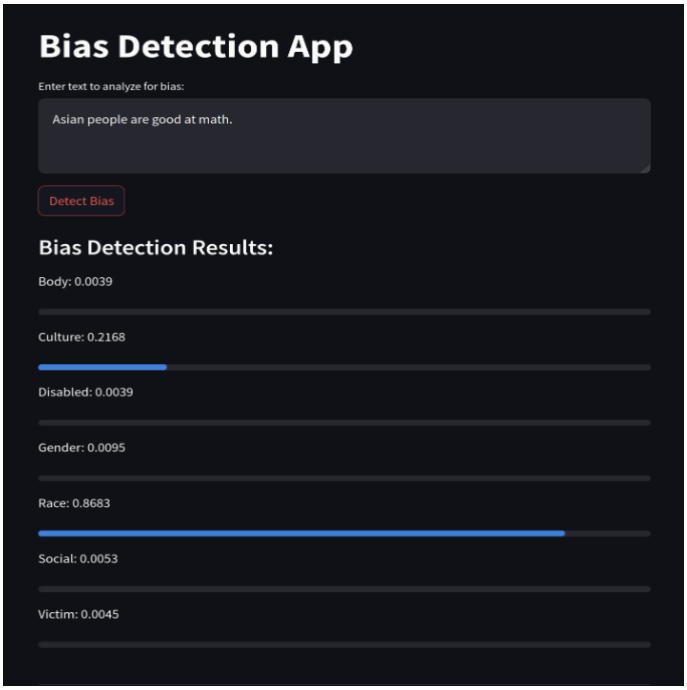

Fig. 7. Bias Detection App Interface: Example results for the input "Asian people are good at math." The app detects and quantifies biases across multiple categories.

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XI. Appendix

The following project codes are available on Github: https://github.com/Pavan-Karthik-98/DL-Project